

SANTUSHT KOTAI

SOFTWARE ENGINEER | BACKEND ENGINEERING | DISTRIBUTED SYSTEMS

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SUMMARY

Backend engineer with hands-on experience building REST APIs, distributed systems, and cloud-native applications in Python, FastAPI, PostgreSQL, Redis, Docker, and Kubernetes. Background spans a production backend internship, an active Google Summer of Code contribution, and independent systems projects covering authentication, caching, and mail infrastructure.

CORE EXPERTISE

- Backend Engineering: REST APIs, Microservices Architecture, gRPC, WebSockets, JSON, Web Services, OpenAPI/Swagger
- Cloud and DevOps: AWS (IaaS, PaaS, SaaS), Docker, Kubernetes, CI/CD, Terraform, GitHub Actions, Jenkins
- System Design: Distributed Systems, Event-Driven Architecture, Horizontal Scaling, Fault Tolerance, High Availability
- Security: OAuth 2.0, JWT, RBAC, OWASP Top 10, API Rate Limiting, AES Encryption, TLS/SSL

SKILLS

Languages: Python, Golang, Java, C++, C, JavaScript, SQL

Backend: FastAPI, Gin, Spring Boot, Node.js, SQLAlchemy, REST API, OpenAPI/Swagger, gRPC, WebSockets

Cloud/DevOps: AWS (EC2, S3, Lambda, RDS, ECS, API Gateway, SQS, CloudWatch, IAM), Docker, Kubernetes, GitHub Actions

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis - distributed caching, query optimization, indexing

Systems: Client-Server Architecture, Networking Protocols (HTTP, TCP/IP), Microservices, Operating Systems, Linux

Practices: Agile/Scrum, TDD, System Design, Git, Postman, Prometheus, Grafana, OWASP, Code Review

EXPERIENCE

Backend Engineer Intern, COE - 47 Billion - Indore, India

Dec 2025 - April 2026

- Built and maintained 25+ production REST APIs using FastAPI, PostgreSQL, and SQLAlchemy across five enterprise modules, serving 10,000+ daily API requests with consistently low response latency.
- Developed and deployed backend microservices on AWS using cloud-native design principles, supporting scalable workloads and automated deployment pipelines.
- Implemented OAuth 2.0, JWT authentication, RBAC, and API rate limiting while remediating critical security findings through OWASP-aligned security testing.
- Optimized PostgreSQL performance through indexing, query optimization, connection pooling, and Redis caching, improving application responsiveness and reducing database load.
- Containerized backend services using Docker, deployed workloads on Kubernetes, and contributed to GitHub Actions CI/CD pipelines to improve deployment reliability.

LEADERSHIP, OPEN SOURCE & ACHIEVEMENTS

- Google Summer of Code 2026 (GSoC) - Selected Contributor - Supabase
- Buildverse Education Technology Hackathon winner 2026
- Cisco Certified: Introduction to Modern AI, Python Essentials 1 and 2, Introduction to Cybersecurity, Introduction to Networks.
- LeetCode Knight Ranked among the Top 1.2% globally, with 500+ problems solved across Data Structures & Algorithms, including Graphs, Dynamic Programming, Trees, and Advanced Algorithms.
- Founder & Lead Engineer Sharexpress Foundation: Founded an open-source software organization focused on developer infrastructure, distributed systems, and AI-powered platforms. Lead system architecture, technical roadmap, engineering standards, and end-to-end development across multiple open-source products.
- Open Source Contributor - React, Supabase, Vite, FastAPI, linux

PROJECTS

Sharexpress Mail Infrastructure & Cloud Services github.com/santusht06/sharexpress

Jan 2026 - June 2026

FastAPI, React, Redux Toolkit, Cockroach DB, SMTP, IMAP, POP3, Docker, Ngix, Kubernetes. Minio.

- Engineered a self-hosted mail platform from scratch, implementing core mail services using SMTP, IMAP, and POP3 protocols with 30+ REST APIs, RBAC, Redis caching, and an event-driven backend architecture for scalable mail delivery and user management.
- Designed and deployed the complete infrastructure stack, including containerized services, reverse proxy, service orchestration, background workers, secure authentication, and persistent storage. Optimized performance through database indexing, Redis caching, asynchronous processing, and blue-green deployment strategies.

Interleet - AI Interview, Coding & System Design Platform github.com/santusht06/interleet

April 2026 - Present

FastAPI, MongoDB, Redis, LangChain, React 19, Redux Toolkit, Docker, WebSockets, AWS, Open Source Licensing.

- Engineered a distributed AI assessment platform for mock interviews, coding challenges, and system design, featuring multi-provider LLM orchestration, secure authentication, and 35+ REST APIs to deliver scalable, real-time technical assessments.
- Built a secure code execution and evaluation engine using non-root Docker sandboxes, AST-based analysis, differential testing, mutation testing, and Redis-backed asynchronous processing to safely execute and accurately validate untrusted code.

EDUCATION

B.Tech - Computer Science and Engineering | Medicaps University, Indore, India

Aug 2023 - Aug 2027

Coursework: DSA, Operating Systems, Computer Networks, DBMS, System Design, Distributed Systems, Cloud Computing, Software Engineering